

Manual Automatic Map Production

This is the master manual for turning coordinate data into publication-ready maps.

Document Metadata

- Audience: SOPD operators, GIS staff, and community contributors
- Scope: automated map production from coordinate data
- Output types: PNG, PDF, SVG, QGIS project, ISO 19139 metadata
- Cartographic base: kabupaten, kecamatan, desa

Use This When

- You have CSV, GeoJSON, VRT, or GeoPackage coordinate data
- You need BIG / PALAPA-ready outputs
- You need PNG, PDF, SVG, metadata, and a reusable QGIS project
- You want SOPD teams to follow one consistent workflow

Fast Path

If you only need the shortest possible workflow, start here: - docs/guides/sopd-quick-start.md

Core Rules

- Base map: kabupaten, kecamatan, desa
- Prefer GeoPackage over shapefile
- Keep thematic data separate from the base map
- Use a sector profile from `shared/config/sector_profiles.json`
- Warn users that same-named outputs will be overwritten

Minimal Steps

1. Put source data in `projects/<project>/data/`.
2. Create or update the project GeoPackage at
`projects/<project>/data/<project>.gpkg`.
3. Generate metadata.

4. Render the map.
5. Check outputs in `projects/<project>/output/`.

Validation Checklist

- CSV or GeoJSON coordinates are valid
- `longitude / latitude` are populated and correct
- Base map layers are present
- Metadata XML parses successfully
- QGIS project uses relative paths
- Output files were created in `output/`
- Same-named outputs were intentionally overwritten

Commands

Generate metadata

```
python3 shared/scripts/generate_big_metadata.py \  
  
--template shared/metadata/batas_admin_metadata.xml \  
  
--output projects/<project>/metadata/<dataset>_metadata.xml \  
  
--title "<Title>" \  
  
--abstract "<Abstract>" \  
  
--source "<Source>" \  
  
--bbox "xmin,ymin,xmax,ymax"
```

Render maps

```
python3 shared/scripts/generate_professional_map.py \  
  
--project projects/<project> \  
  
--sector <sector-name> \  
  
--layers <thematic-layer> \  
  
--output-format png,pdf,svg \  
  
--dpi 300
```

Outputs

- output/*.png
- output/*.pdf
- output/*.svg
- qgis/*.qgs
- data/<project>.gpkg when the project is packaged as a self-contained GeoPackage

PDF / Print Export Note

This markdown file is intended to be print-friendly.

To generate a PDF version, use your preferred Markdown-to-PDF tool, for example:

```
pandoc docs/guides/manual-automatic-map-production.md -o manual-  
automatic-map-production.pdf
```

Recommended Sector Profiles

- admin
- health
- infrastructure
- education
- public_works
- agriculture

Metadata Must Include

- UUID file identifier
- `gmd:metadataExtensionInfo`
- `gmd:useLimitation`
- lineage
- source description
- contact information

Troubleshooting

- If a map looks empty, verify the input coordinates and CRS.
- If the base map is missing, confirm `kabupaten`, `kecamatan`, and `desa` exist in the project GeoPackage or shared shapefiles.
- If the wrong outputs appear, check for same-named files in `projects/<project>/output/`.
- If metadata fails to parse, validate the XML before upload.
- If the project is `faskes-kobar`, the rebuilt GeoPackage should contain 94 villages and 10 faskes points.

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